25900682 MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2025

THIRD SEMESTER
DISCIPLINE SPECIFIC CORE COURSE (DSC)-MG3DSCZGY200, MG3DSCZIM200

PROTISTAN DIVERSITY AND ANIMAL DIVERSITY – NON CHORDATA - I
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions. Answer all questions.
Each question carries **1** marks **[1x12=12]**

1. The mode of transmission of *Ancylostoma duodenale* : [K] [CO4]
(b) Skin penetration

2. Identify the class to which Obelia belongs: [K]/[CO3]
(a) Hydrozoan

3. Diatoms are often found as ---------- [K]/[CO1]
(b) Brown slippery coating on submerged stones

4. Which stage of Plasmodium is infectious to humans [K]/[CO1]
(d) Sporozoite

5. How can the spread of amoebic dysentery be prevented? [U]/[CO1]
(c) Maintaining hygiene and using clean drinking water

6. Body symmetry of Porifera is ---------- [K]/[CO2]
(a) Radial/asymmetrical -

7. Ascaris lumbricoides belongs to which class of nematodes? [K]/[CO4]
(c) Phasmida

8. What is a distinctive morphological feature of Trichonympha? [K]/[CO1]
(c) A large number of flagellacovering the body

9. In Protista, locomotion is affected by ---------- [K]/[CO1]
(d) All of the above

10. Which class of annelids shows ectoparasitic mode of life? [K]/[CO4]
(a) Hirudinea

11. *Giardia intestinalis* belongs to: [K]/[CO1]
(a) Phylum Metamonada.

12. Locomotion in Paramecium is carried out by …… [K]/[CO1]

(b). Cilia

PART B

Answer any 6 questions.
Each question carries 3 marks [3x6=18]

13. Write an account on the adaptations of nematodes for parasitic life. [U]/[CO4]

Body is covered by a thick, flexible, and resistant cuticle that protects them from digestive enzymes and host immune responses. The elongated, cylindrical, and tapering body helps them move easily through host tissues. Hooks, suckers, or lips around the mouth for firm attachment to host tissues (eg.*Ancylostoma* has cutting plates). Produce a large number of eggs to ensure survival of offspring. Eggs and larval stages have thick protective shells to withstand unfavourable conditions. Show reduction in sensory organs. Simplified digestive tracts suited to absorbing pre-digested host nutrients. They can survive in oxygen-poor environments of the host intestine through anaerobic metabolism.

(Any 3 points -3Marks)

14. Describe two structural features of the test of Foraminifera. [K]/[CO1]

The test is usually calcareous (made of calcium carbonate). It is divided into a series of chambers (loculi) and come in shapes like spirals or globules. The test wall has many minute pores (foramina) through which slender extensions of the protoplasm (pseudopodia) project for feeding and locomotion.

(Any 2 points -1.5marks each (Liberal valuation))

15. Discuss on the economic importance of Red algae. [U]/[CO1]

It is a source of agar: used in laboratories for culture media and in food industries as a gelling and stabilizing agent. **Carrageenan production:** used as a thickener in ice cream, toothpaste etc. **Food Source:** edible and rich in proteins, vitamins, and minerals. **Ecological Role:** They contribute to coral reef formation (e.g., *Corallina*), providing habitat and stability to marine ecosystems. Used in pharmaceuticals and cosmetics, Used as culture medium in microbiology, Used in textile and leather industries, Used as fertilizers and soil conditioners, Source of iodine and other minerals, Used in paper and glue making.

(Any 3 points -3marks)

16. Out line the public health significance of *Schistosoma haematobium.* [U]/[CO4]

It is a blood fluke that causes **urinary schistosomiasis.** Common in Africa, the Middle East, and parts of India. Poor sanitation and unsafe water promote its spread. It affects mainly in **children and agricultural workers** exposed to contaminated water. Associated with **malnutrition and anaemia** in chronic cases.

(Any 3 points -3marks)

17. Differentiate the Syconoid canal system from the Asconoid type in sponges. [U]/[CO2]

Feature	**Asconoid Canal System**	**Syconoid Canal System**
1. Complexity	Simplest type of canal system	More complex than asconoid
2. Body Wall	Thin and not folded	Body wall is folded into finger-like projections called *radial canals*
3. Canals	No radial canals; water flows directly through the spongocoel	Presence of radial canals lined with choanocytes
4. Flow of Water	Ostia → Spongocoel → Osculum	Ostia → Incurrent canal → Prosopyles → Radial canal → Apopyle → Spongocoel → Osculum
5.Choanocytes	Line the spongocoel	Line the radial canals, not the

(Collar Cells)		spongocoel
6.Efficiency of Circulation	Less efficient due to small surface area for filtering	More efficient as surface area for filtering is increased
7. Example	*Leucosolenia*	*Sycon* (also called *Scypha*)

(Comparison of any 3 points -3marks, No need of diagram. If students draw diagram only give 1.5 marks)

18. Describe the organ system level of organization in animals. [U]/[CO1]

Organ system level of organization means the body of an animal is made up of different organs that work together to perform specific functions. Each organ is made of tissues. E.g.: digestive system, respiratory system etc. This type of organization is seen in more advanced multicellular animals such as annelids, arthropods, molluscs, and vertebrates. It marks a high level of specialization and division of labour among the organs and systems.

(Give full marks for relevant points)

19. Define alloiogenesis and explain it with a suitable example. [U]/[CO4]

Alloiogenesis in Platyhelminthes refers to the alternation of sexual and asexual generations in the life cycle of certain parasitic flatworms. In the definitive host (usually a vertebrate), the adult worm reproduces sexually, producing fertilized eggs that are released into the environment. In the intermediate host (often a snail or invertebrate), the larval stages such as sporocyst and redia multiply asexually, increasing the number of infective larvae.

Eg: In Schistosoma or Fasciola hepatica (liver fluke).

(Liberal valuation)

20. Classify corals under their respective phylum and outline two distinguishing traits. [U]/[CO3]

Corals belong to the Phylum Cnidaria and Class Anthozoa. They are exclusively marine, sessile organisms that exist either singly or in colonies. Corals are divided mainly into hard corals (order Scleractinia) and soft corals (order Alcyonacea) based on the presence or absence of a hard skeleton.

(Classification 1 mark; any2 traits 1 mark each)

Part C

Short Essay Questions Answer any **4**questions;
Each question carries **5** marks

21. Explain the locomotion, nutrition, and reproduction in Opalina. [U]/[CO1]

Locomotion Opalina moves by coordinated beating of numerous cilia covering its body surface. Cilia beat in a wave-like manner, producing smooth gliding movement. Movement helps it mix with intestinal contents and maintain position in host. No specialized organ for attachment or creeping. ***(1½ marks)***

Nutrition: Holozoic and saprozoic mode of nutrition. Feeds on intestinal debris, bacteria, and organic matter from host intestine. No cytostome (mouth); food is absorbed directly through the body surface by diffusion or phagocytosis. Digestion is intracellular and nutrients are distributed throughout cytoplasm. ***(1½ marks)***

Reproduction: Asexual reproduction: By binary fission (transverse division) producing daughter individuals. **Sexual reproduction:** By syngamy – gametes fuse to form zygote. Encystment occurs

during adverse conditions or before leaving the host. Cysts help in transmission to a new host when swallowed by tadpoles. ***(2 marks)***

(a good account of Opalin -Give full marks)

22. Write short notes on the structure and functions of combplates and colloblasts. [U]/[CO3]

- **Comb Plates (2½ marks):**

Structure (1½ marks): Present in **phylum Ctenophora** (comb jellies). Formed of **fused cilia** arranged in **eight vertical rows** called **ctenes**. Each row functions as a **locomotory organ**, forming a comb-like plate.

Functions (1 mark): Locomotion – propel the animal smoothly through water. **Maintain balance and direction** during movement. Help in **spreading body currents** for respiration.

- **Colloblasts (2½ marks)**

Structure (1½ marks): Unique adhesive cells found on **tentacles of ctenophores**. Each colloblast is a **straight filament**, and a **spiral filament**. The sticky material helps in trapping prey. Nematocyst absent.

Functions (1 mark): Used for **capturing food** by adhesion (not by stinging). Help in **holding and immobilizing prey** before ingestion.

(Liberal valuation)

23. Explain the significance of conjugation. [U]/[CO1]

Conjugation is a type of sexual process seen in *Paramecium*, where two individuals temporarily unite to exchange nuclear material. ***(0.5 marks)***

Rejuvenation: Conjugation restores vitality in organisms that have become weak due to repeated asexual reproduction. It rejuvenates the cell by mixing nuclear material from two individuals, preventing degeneration. **Nuclear Reorganization*:*** Exchange and fusion of micronuclei lead to reorganization of genetic material. This ensures normal nuclear function and maintains genetic stability. **Hereditary Variation**: Conjugation introduces genetic variation by recombining genes from two parent cells. Such variation helps in adaptation and evolution of the species. ***(Explanation of 3 points -1.5 marks each)***

(NB: Please Give 2 mark for explanation of conjugation and 1.5marks for any 2 significance)

24. Discuss the burrowing adaptations of annelids with special reference to Polygordius.

[U]/[CO4]

PolygordiusLives in **sandy and muddy bottoms**, showing complete adaptation to a **burrowing, tube-dwelling lifestyle**.

1) **Body Shape:** The body of *Polygordius* is **elongated, slender, and cylindrical**, allowing easy penetration through sand and mud.
2) **Segmentation:** The body is **metamerically segmented**, providing flexibility and coordinated movement during burrowing.
3) **Setae and Parapodia:** Setae and **parapodia areabsent**, minimizing friction and aiding smooth forward and backward movement in the substrate.
4) **Cuticle:** The **thin, moist cuticle** secretes mucus that **reduces friction** and helps the worm glide through sediment.
5) **Cilia:** Cilia on the epidermis help in **movement of sand particles** and cleaning of the body

surface.

6) **Head and Sensory Organs:** The head is **poorly developed and without eyes or antennae**, reducing drag and helping it remain protected while burrowing.
7) **Skin and Respiration:Thin, vascularized skin** enables **cutaneous respiration**, effective in oxygen-poor environments.
8) **Circulatory System:** A **closed circulatory system** efficiently transports oxygen and nutrients even under low-oxygen burrowing conditions.
9) **Musculature:Well-developed circular and longitudinal muscles** provide powerful contractions for digging.

(Explanation of any 5 points -5 marks or mention all points 5 marks)

25. Discuss the types of coelom and explain their significance in the classification of animals.
[U]/[CO2]

Coelom is the **body cavity** between the body wall and gut wall, lined by **mesoderm**. It provides space for organ development and movement. Coelom is an important **basis for classifying animals.** ***(Definition 0.5 mark)***

Types of Coelom: 1)Acoelomate: No body cavity between body wall (ectoderm) and gut (endoderm); space filled with mesenchyme. E.g.: Flatworms (*Platyhelminthes*). 2) **Pseudocoelomate:** Body cavity present but not fully lined by mesoderm; derived from blastocoel. E.g.:Roundworms (*Ascaris, Nematoda*). 3) **Eucoelomate (True Coelomate):** Body cavity fully lined by mesoderm forming peritoneum; allows organ specialization.e.g.: Earthworm (*Annelida*), Molluscs, Arthropods, Chordates.

(3 Types of Coelom with example / labelled diagrame with explanation -1.5 marks each)

26. Discuss the parasitic adaptations of Phylum Platyhelminthes. [U]/[CO4]

1. Presence of Suckers and Hooks (Major Point): Powerful suckers (oral and ventral) help in firm attachment to host tissues. Hooks on the scolex of *Taenia solium* assist in anchoring to the intestinal wall. *Fasciola hepatica* has anterior and ventral suckers for clinging to bile ducts.
2. Protective Tegument: Thick, resistant cuticle or tegument protects the parasite from host digestive enzymes and immune reactions (*Taenia*, *Schistosoma*).
3. Reduced Digestive System: Parasitic forms absorb pre-digested food directly through the tegument; gut may be simple or absent (*Taenia* lacks an alimentary canal).
4. High Reproductive Capacity: Hermaphroditism and numerous eggs ensure survival and continuation of the species. Alloiogenesis in Platyhelminthes refers to the alternation of sexual and asexual generations in the life cycle of certain parasitic flatworms.
5. Complex Life Cycle with Intermediate Hosts: Multiple larval stages and alternation of hosts increase chances of transmission .
6. Well-developed Excretory and Nervous Systems: Flame cells remove waste efficiently; simple nervous system coordinates attachment and movement.
7. Reduction of Non-essential Organs: Sense organs and locomotory organs are reduced or absent as parasites live in stable environments.
8. Adapted Mode of Respiration: Anaerobic respiration allows survival in oxygen-poor internal environments of the host.
9. Encystment Ability: Larval stages can encyst and remain dormant until reaching a suitable host (*Echinococcus* hydatid cysts)

(Explanation of any 5 points -5 marks or mention all points 5 marks)